# Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 08

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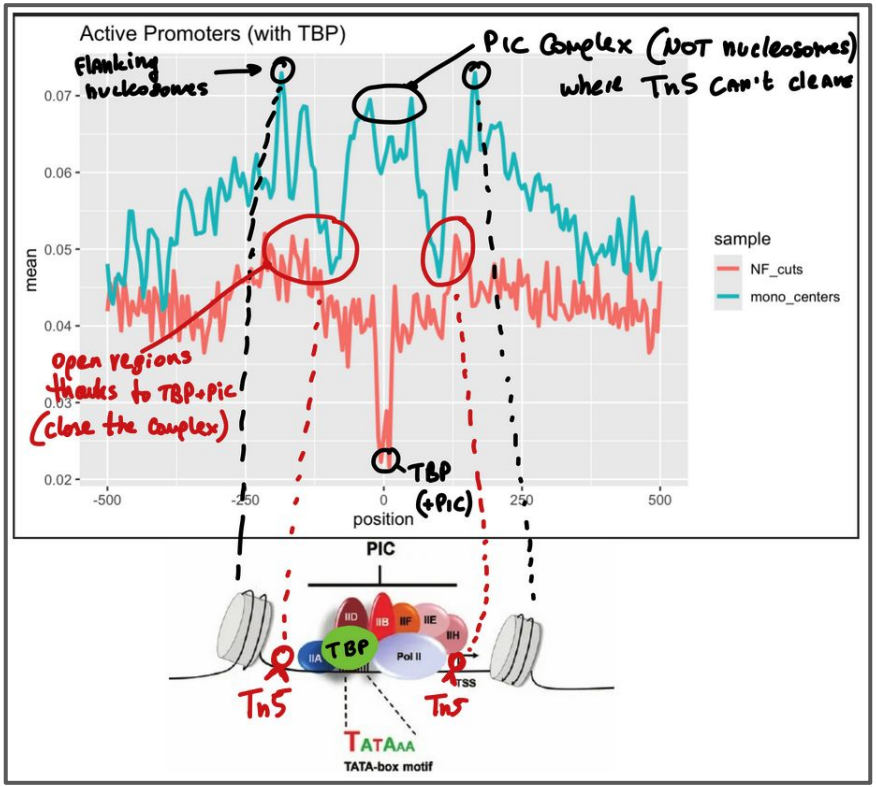
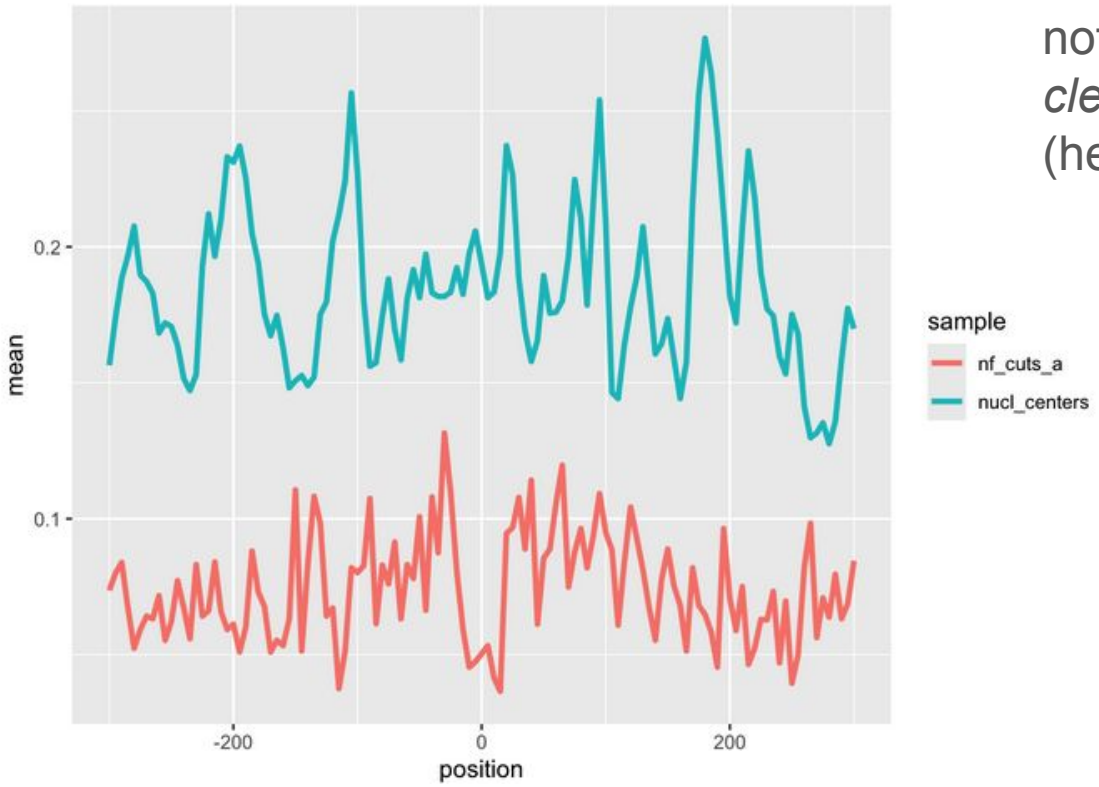
# Plan for Today

- Debriefing on the assignment
- Some biology behind DNA Methylation
- Practical 1: visualization of DNA methylation
- Profiling Techniques
- Practical 2: statistical analysis

# Debriefing on the assignment

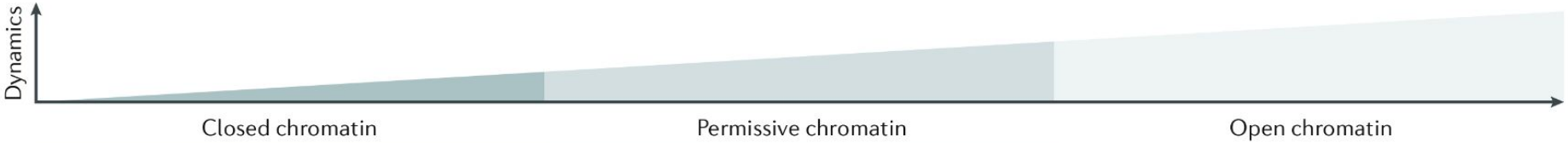


# Debriefing on the assignment




not all TFs leave *clear* footprints (here ATF3)

# Degrees of activation/repression




**Heterochromatin (stable repression)**

Very low accessibility

**DNA methylation**

HP1 binding

H3K9 methylation

**Poised/silenced**

Low accessibility

H3K27 methylation

**Active elements**

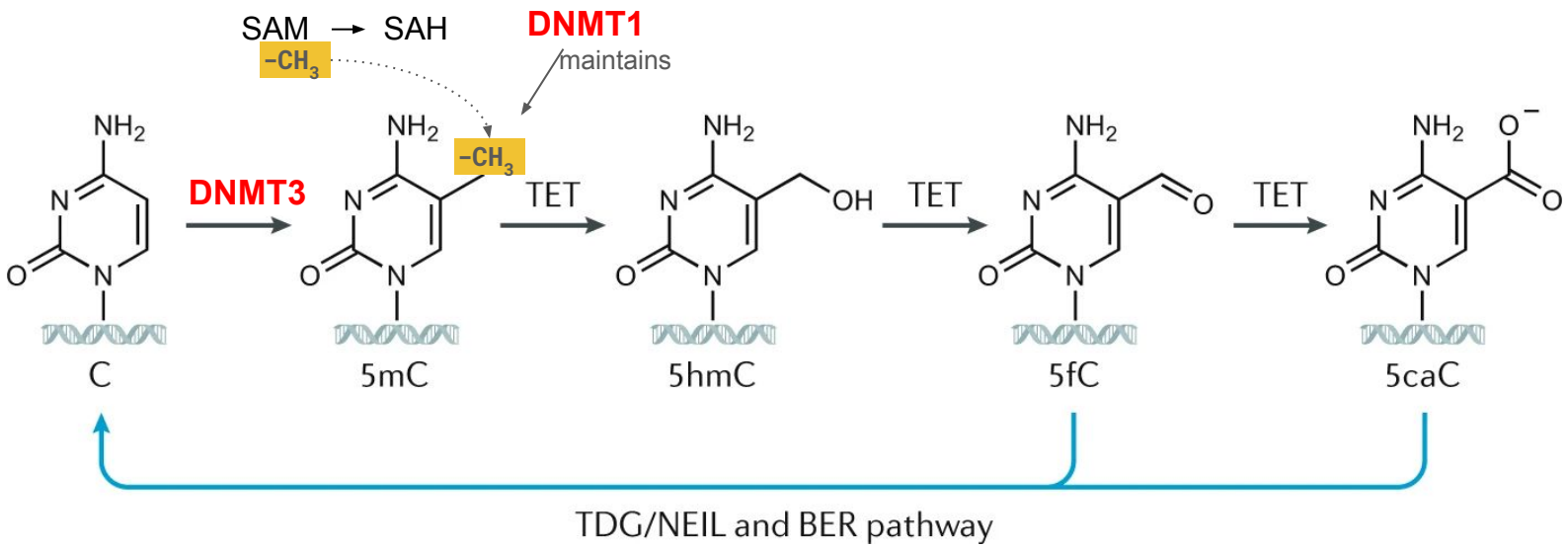
Very high accessibility

RNApol2 binding
H3K4 methylation
H3K27 acetylation

Fate transitions
(e.g. differentiation)

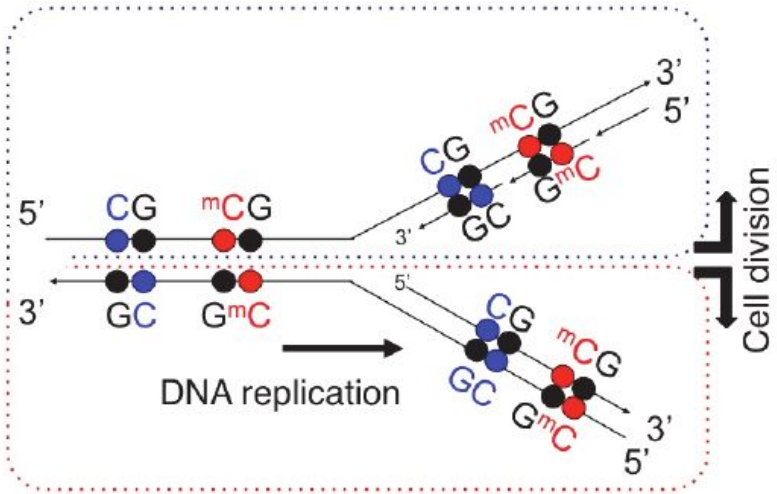
Transient responses

# What is DNA methylation?




(Modified from Parry, Rulands and Reik, *Nat Rev Gen* 2021)

# Methylation context - why all this talk about CpGs ?




(Adapted from Kubota et al., 2011)

At C-G pairs, DNMT1 ensures that the methylation is copied to the newly synthesized DNA strands

While methylation does occur at non-CpGs, it normally gets diluted out during replication

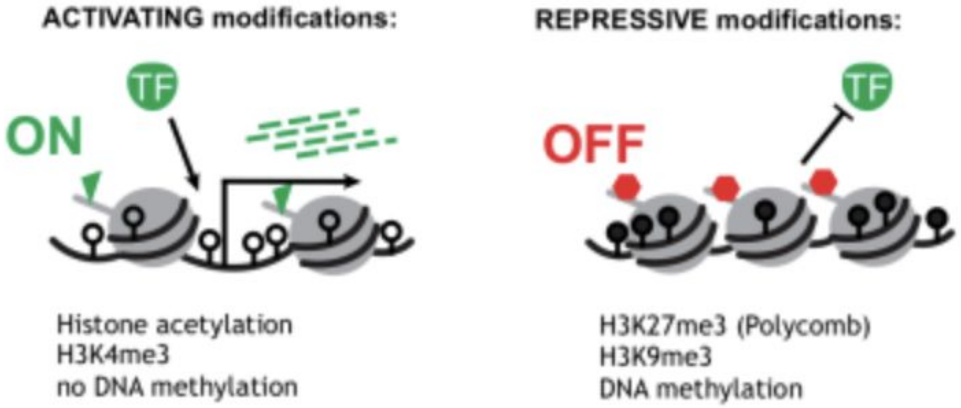
Hence (in mammals) most relevant DNA methylation happens at CpGs.

<u>Some exceptions:</u>

- In embryonic stem cells
- In post-mitotic neurons
- In cancer

# DNA Methylation: Activation vs Repression

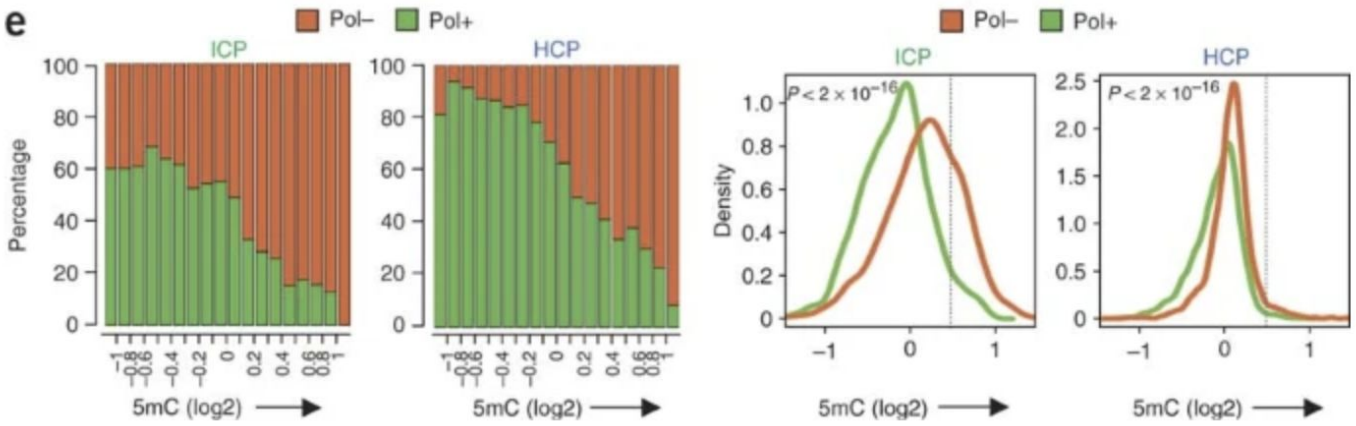
Repressive role of DNA methylation in transcription




Scheme from Izaskun Mallona/Tuncay Baubec

Correlation between DNA methylation and gene silencing

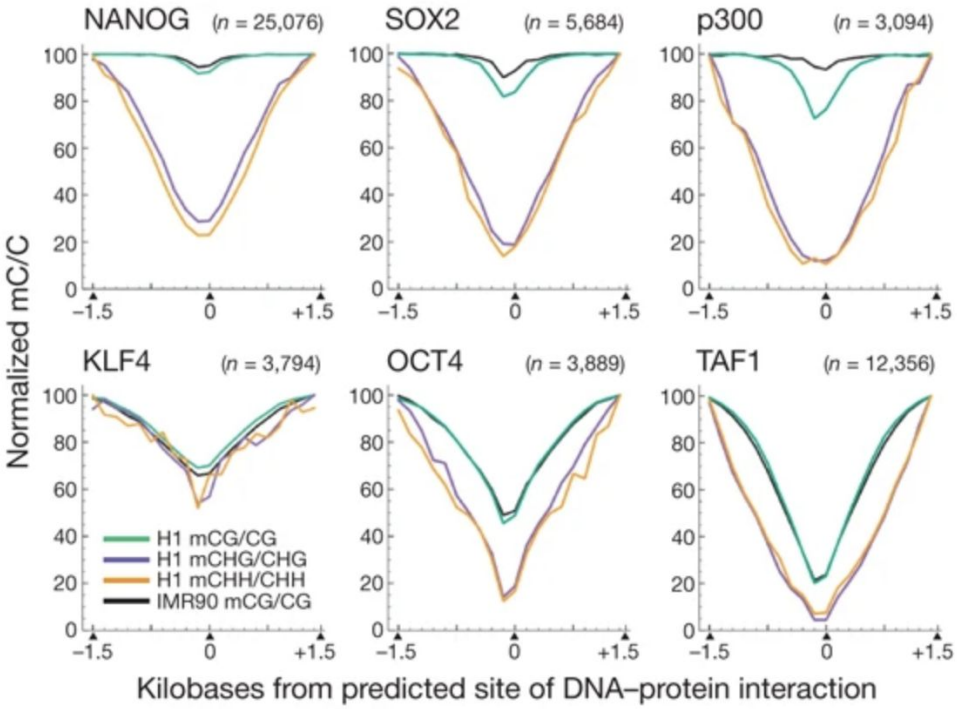
# At promoters, DNA methylation has long been associated to repression




(adapted from Weber et al., 2007)

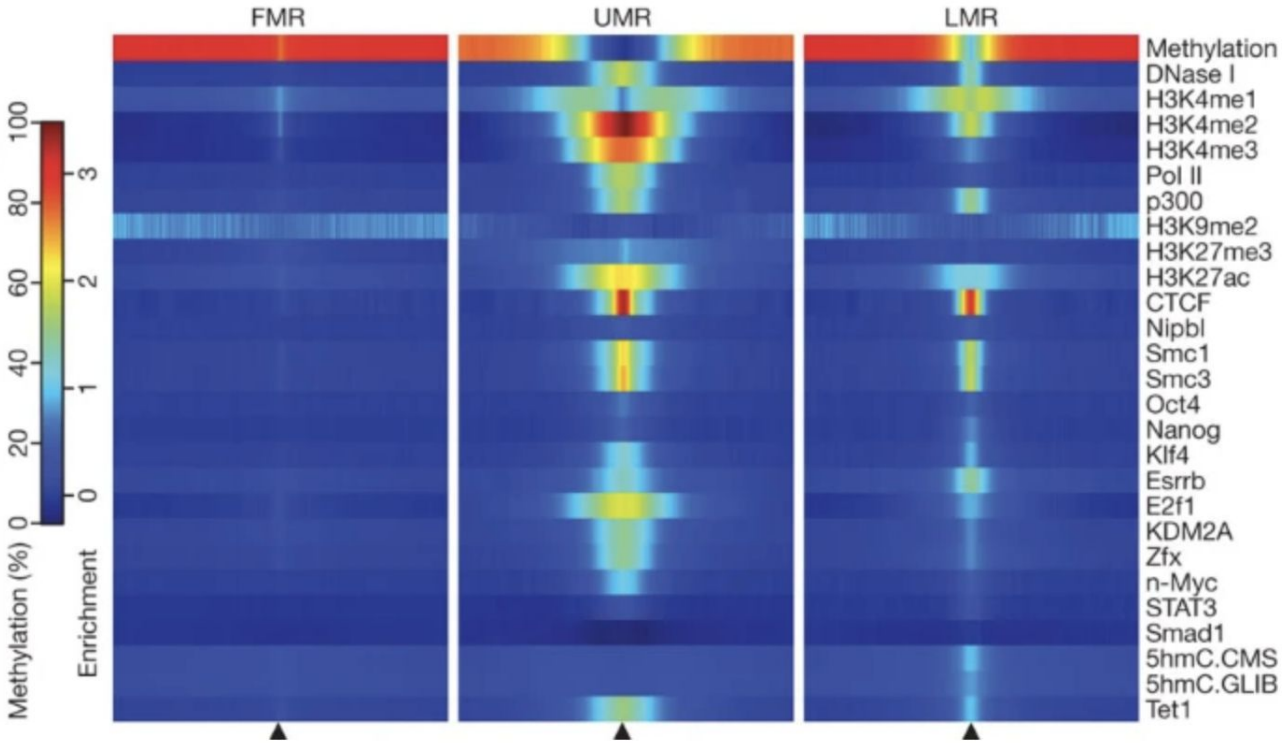
We now know that this repression is mostly happening because the methylation inhibits the binding of transcription factors (TFs) (Kaluscha et al., Nat Gen 2022)

# TF Binding & Methylation




(adapted from Lister et al., 2009)

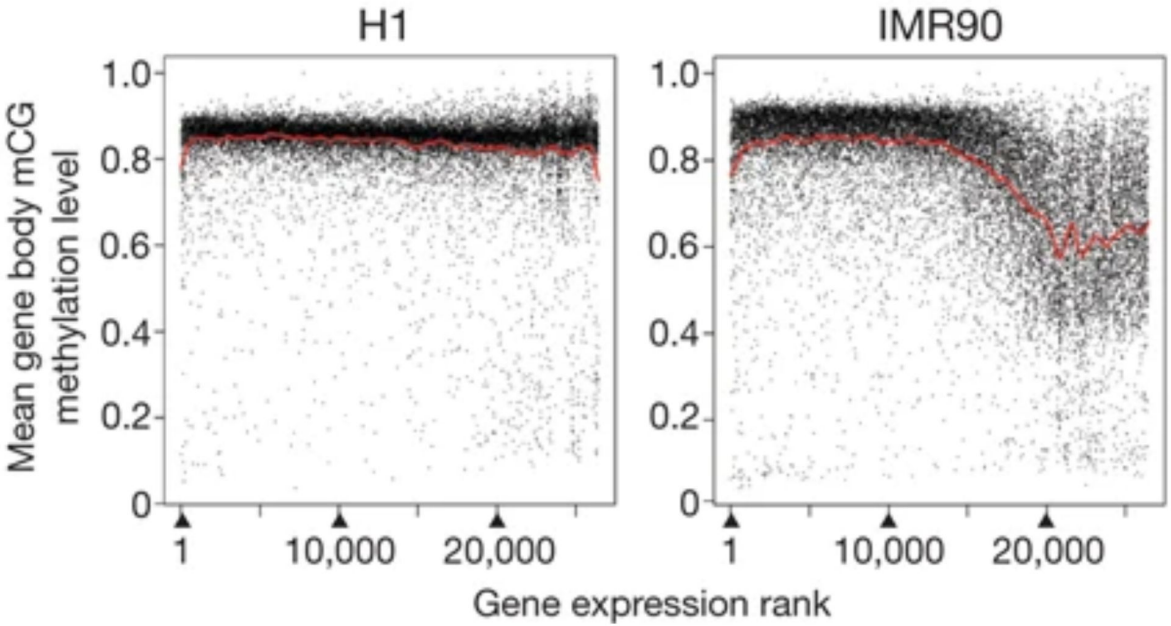
# TF Binding & Methylation




(Adapted from Stadler et al., 2011)

# DNA methylation of gene bodies

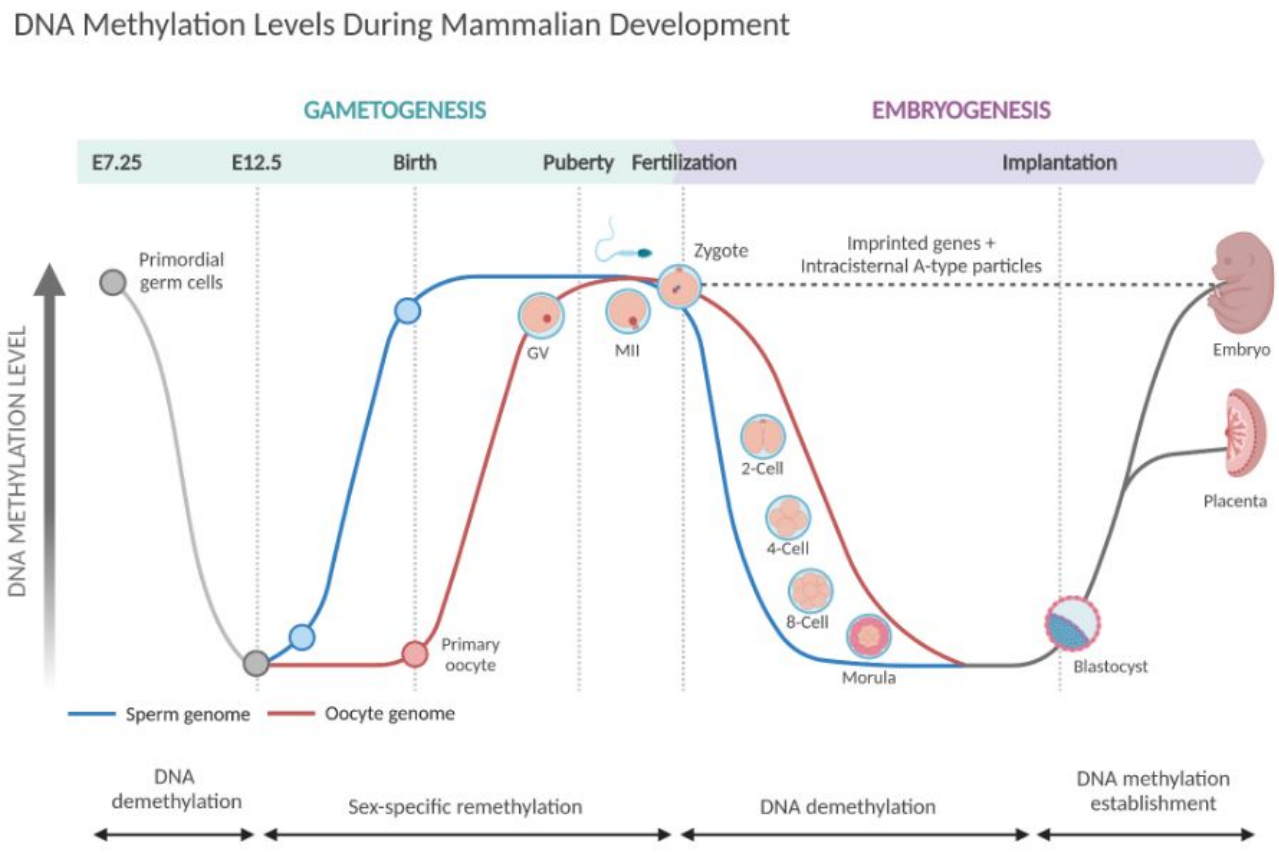
*"The enrichment of DNA methylation in gene bodies presents a paradox: on the one hand, gene-body methylation is highly conserved across eukaryotes — more than it is conserved at transposable elements, for example — indicating it has an important function. On the other hand, DNA methylation is mutagenic, so why is it so prominent in coding sequences" (Greenberg et al. 2019)*




Differentiated cell types typically show a positive correlation between gene expression and gene body methylation

Lister et al., 2009

# Germline DNA methylation reprogramming



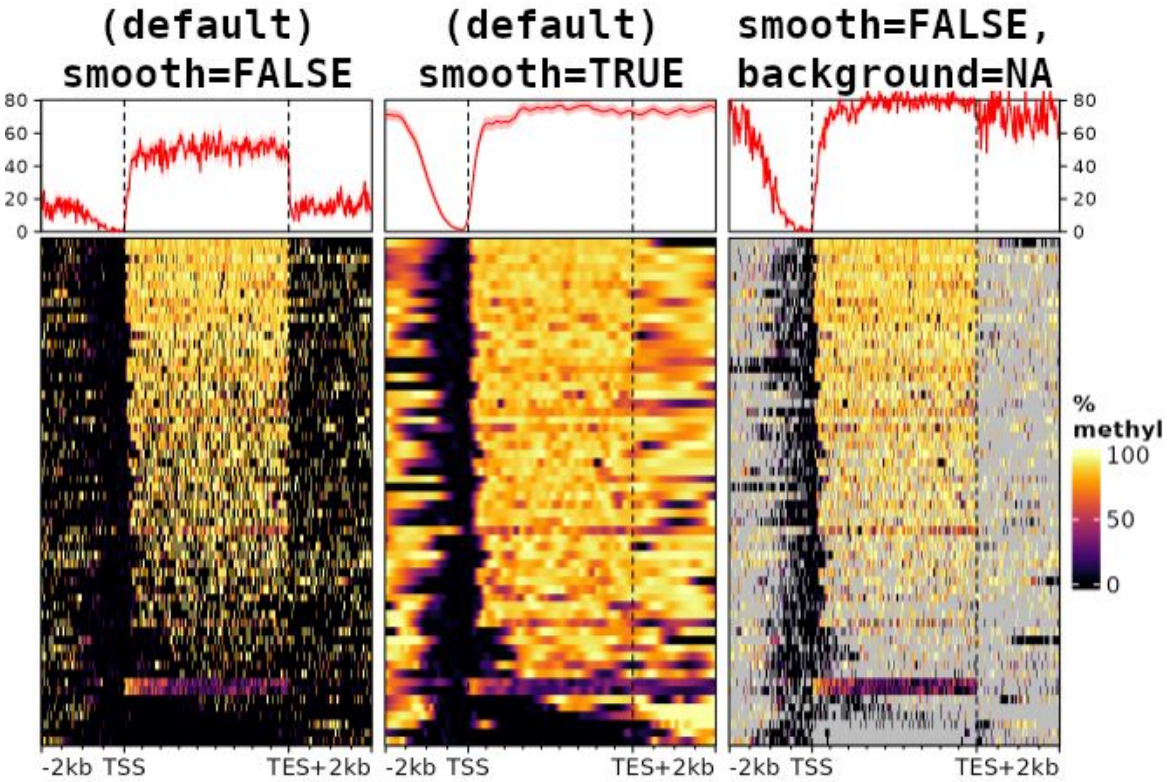

(BioRender)

# Practical - Part 1

Visualization

# Visualization of DNAme data in epiwraps

DNAme differs from other signals in that there can be an absence of signal not simply because the modification is absent, but because there aren't any C's to be methylated…



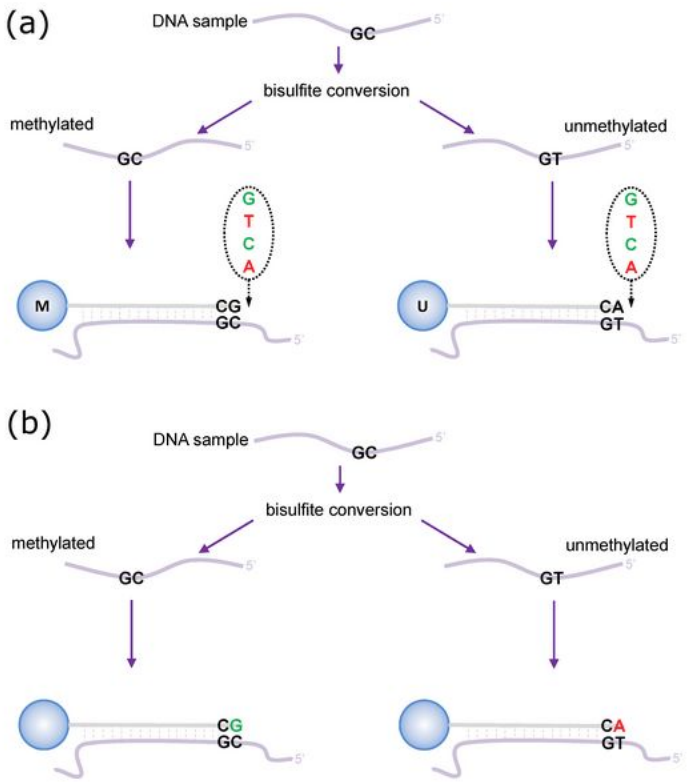

Gene bodies in a cancer cell line

See the [epiwraps documentation]

# Profiling techniques

- DNA Methylation Arrays:
  - Illumina 450K/850K
- Bisulfite sequencing:
  - RRBS
  - WGBS
  - scWGBS

# Methylation Arrays




- fluorescence signal
- cheap(er)
- again Illumina:
  - Infinium 450K & 850K probing wide ranges of the genome

- Analysis:
  - [A cross-package Bioconductor workflow for analysing methylation array data.]

Maksimovic et al., 2016

# Beta & M-Values

- Beta-value, ratio of the methylated probe intensity and the overall intensity:

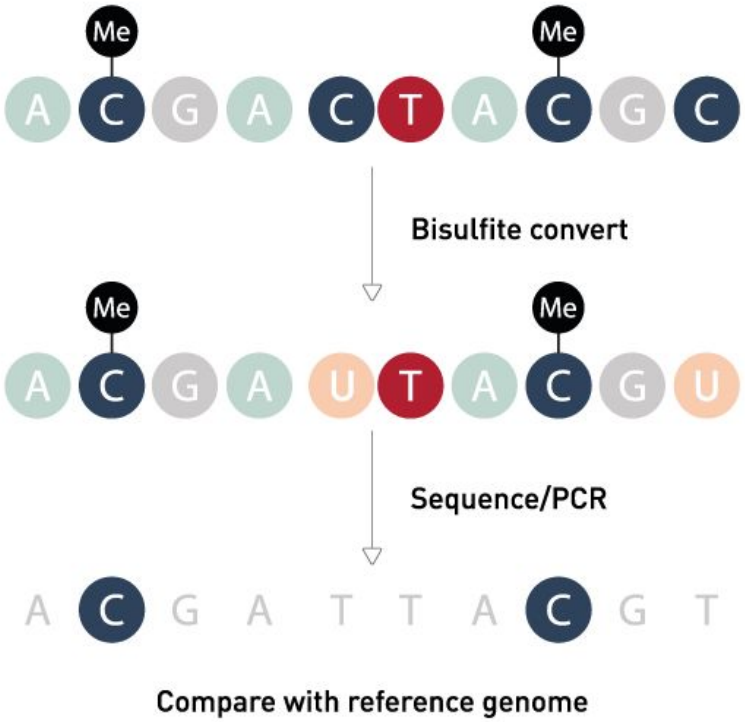
$$Beta_i = \frac{\max(y_{i,methy}, 0)}{\max(y_{i,unmethy}, 0) + \max(y_{i,methy}, 0) + \alpha}$$

- M-value, log2 ratio of the intensities of methylated probe versus unmethylated probe:

$$M_i = \log_2 \left( \frac{\max(y_{i,methy}, 0) + \alpha}{\max(y_{i,unmethy}, 0) + \alpha} \right)$$

- The Beta-value has a more intuitive biological interpretation, but the M-value is more statistically valid for the differential analysis of methylation levels. (Du et al., 2010)

# Bisulfite Sequencing




https://www.diagenode.com/en/categories/bisulfite-conversion

Exists in different flavors:

- RRBS
- WGBS
- scWGBS

# Bisulfite Sequencing - Aligning

## Bismark

CpG-report file:

| | chr | pos | strand | met | unmet | context | tri-context |
|---|---|---|---|---|---|---|---|
| 1: | 5 | 11823 | + | 0 | 0 | CG | CGA |
| 2: | 5 | 11824 | - | 0 | 0 | CG | CGG |
| 3: | 5 | 11850 | + | 0 | 0 | CG | CGA |
| 4: | 5 | 11851 | - | 0 | 0 | CG | CGG |
| 5: | 5 | 11856 | + | 0 | 0 | CG | CGA |

In R: bsseq package:

```
read.bismark()
```

# Bisulfite Sequencing - Aligning

## Bismark

CpG-report file:

| met | unmet | |
|---|---|---|
| 0 | 0 | |
| 20 | 20 | |
| 1 | 2 | → 50% |
| 10 | 20 | → 50% |

| | chr | pos | strand | met | unmet | c | |
|---|---|---|---|---|---|---|---|
| 1: | 5 | 11823 | + | 0 | 0 | | |
| 2: | 5 | 11824 | - | 0 | 0 | CG | CGG |
| 3: | 5 | 11850 | + | 0 | 0 | CG | CGA |
| 4: | 5 | 11851 | - | 0 | 0 | CG | CGG |
| 5: | 5 | 11856 | + | 0 | 0 | CG | CGA |

In R: bsseq package:

```
read.bismark()
```

# DNA-Methylation Analysis tasks:

- Smoothing:
  - bsseq
- Differential Methylation testing:
  - DMRcate (defines DM regions itself)
  - edgeR (allows for testing user-defined regions, e.g. genes of interest)
  - Many more (Methylkit, bsseq etc.)
- Calling partially methylated domains (PMDs)
  - MethylSeekR
- Quantifying DNA methylation heterogeneity
  - Quantitative comparison of within-sample heterogeneity scores for DNA methylation (Scherer et al., 2020)

For more details regarding differential methylation testing see: Chen et al, 2018, https://f1000research.com/articles/6-2055/v2

# Practical - Part 2

bsseq objects + Differential Methylation Testing

# Assignment

**1.** Plot a heatmap of the methylation levels of the genes in top 5 DMR regions (from the practical) located on chr1. You may either do this as we did for promoters during the practical (i.e. using `getMeth()`), or using `signal2Matrix()` and `plotEnrichedHeatmaps()`).

**2.** Use the differentially-methylated regions from the practical, and compare those changes to changes in accessibility. For the latter, you can download differential accessibility results from here:

https://www.ethz-ins.org/content/ATAC_AstroVsESC_subset.rds

Given these two datasets, figure out a way (visualization/analysis) of assessing the extent of agreement (or disagreement) between changes in these two modalities. Describe what you did and what it means in a sentence or two.

# References


- Weber, M., Hellmann, I., Stadler, M. et al. Distribution, silencing potential and evolutionary impact of promoter DNA methylation in the human genome. Nat Genet 39, 457–466 (2007). https://doi.org/10.1038/ng1990
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- Kaluscha, S., Domcke, S., Wirbelauer, C. *et al.* Evidence that direct inhibition of transcription factor binding is the prevailing mode of gene and repeat repression by DNA methylation. *Nat Genet* 54, 1895–1906 (2022). https://doi.org/10.1038/s41588-022-01241-6